The Invariance Principle — problem 1.21

A. Engel — Problem-Solving Strategies · 1998

Three integers a, b, c are written on a blackboard. Then one of the integers is erased and replaced by the sum of the other two diminished by 1. This operation is repeated many times with the final result $17, 1967, 1983$. Could the initial numbers be (a) $2, 2, 2$ (b) $3, 3, 3$?